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Practical 2

Aim : Design a distributed application using RMI for remote computation where client submits two strings to the server and server returns the concatenation of the given strings.

Input :

Server code :

```
import java.rmi.server.UnicastRemoteObject;
import java.rmi.RemoteException;
import java.rmi.registry.LocateRegistry;
import java.rmi.Naming;

public class Server extends UnicastRemoteObject implements StringConcat {

    // Constructor required by UnicastRemoteObject
    protected Server() throws RemoteException {
        super();
    }

    // The actual remote computation logic
    @Override
    public String concatStrings(String str1, String str2) throws RemoteException {
        System.out.println("Client requested concatenation for: '" + str1 + "' and '" + str2 + "'");
        return str1 + str2;
    }
}
```

```
public static void main(String[] args) {  
    try {  
        // Start the RMI registry on the default port 1099  
        LocateRegistry.createRegistry(1099);  
  
        // Create an instance of the server  
        Server server = new Server();  
  
        // Bind the server instance to the registry with a unique name  
        Naming.rebind("rmi://localhost/StringConcatService", server);  
  
        System.out.println("RMI Server is ready and waiting for requests...");  
    } catch (Exception e) {  
        System.err.println("Server exception: " + e.toString());  
        e.printStackTrace();  
    }  
}
```

Client Code :

```
import java.rmi.Naming;
import java.util.Scanner;

public class Client {
    public static void main(String[] args) {
        try {
            // Look up the remote object from the registry
            StringConcat stub = (StringConcat) Naming.lookup("rmi://localhost/StringConcatService");

            Scanner scanner = new Scanner(System.in);

            // Get user input
            System.out.print("Enter the first string: ");
            String string1 = scanner.nextLine();

            System.out.print("Enter the second string: ");
            String string2 = scanner.nextLine();

            // Invoke the remote method
            String result = stub.concatStrings(string1, string2);

            System.out.println("Result from server: " + result);

            scanner.close();
        } catch (Exception e) {
            System.err.println("Client exception: " + e.toString());
            e.printStackTrace();
        }
    }
}
```

```
}  
  
}  
  
}
```

String Concatenation Code :

```
import java.rmi.Remote;  
import java.rmi.RemoteException;  
  
public interface StringConcat extends Remote {  
    // Method signature for concatenating two strings  
    String concatStrings(String str1, String str2) throws RemoteException;  
}
```